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USSD & SMS Server Infrastructure Implementation Proposal

For: Liwiro Microcredit

Prepared by: Infocus Technologies Malawi

Date: October 4, 2025

Validity: 90 Days

Executive Summary

This proposal outlines a comprehensive solution to modernize Liwiro Microcredit's USSD and SMS infrastructure by establishing direct connectivity with mobile network operators (TNM and Airtel). The current setup relies on third-party internet routes, resulting in slow USSD response times and complex troubleshooting procedures involving multiple parties. Our proposed solution eliminates these bottlenecks by implementing direct VPN connections to service providers, establishing dedicated infrastructure, and deploying enterprise-grade gateway solutions.

Total Investment: MWK 8,650,000

Project Duration: 2 Months Or less

Current Challenges

The existing USSD infrastructure at Liwiro Microcredit faces several critical limitations:

Performance Issues: USSD traffic routing through third-party internet routes creates significant latency, degrading user experience and potentially affecting transaction completion rates.

Operational Complexity: The involvement of multiple intermediary parties complicates troubleshooting, extends downtime resolution, and reduces overall system reliability.



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Limited Control: Dependency on external routing providers restricts Liwiro's ability to monitor, optimize, and troubleshoot their own infrastructure effectively.

Scalability Concerns: The current architecture may not support future growth or additional service offerings without substantial modifications.

Proposed Solution

We propose a complete infrastructure overhaul that establishes direct connectivity between Liwiro Microcredit and mobile network operators through dedicated VPN tunnels, supported by robust on-premise infrastructure and professional gateway software.

Key Deliverables

Network Infrastructure Establishment: Design and implementation of secure VPN connections between Liwiro's premises and both TNM and Airtel networks, including local area network setup with configured routers and switches optimized for high-availability telecommunications traffic.

Server Infrastructure Deployment: Complete server configuration and hardening to host gateway applications, including operating system optimization, security configurations, and performance tuning specific to USSD and SMS workloads.

Gateway Software Implementation: Supply, installation, and configuration of enterprise SMPP (Short Message Peer-to-Peer) software for SMS handling and USSD TruRoute gateway for direct USSD session management.

Connectivity Validation: Comprehensive testing protocols including ping tests, telnet connectivity verification, latency measurements, throughput testing, and failover scenario validation.

Service Provider Coordination: Complete liaison services with TNM and Airtel, managing all technical discussions, documentation requirements, configuration exchanges, and integration testing schedules.



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Knowledge Transfer: Structured technical handover with comprehensive documentation, training sessions for Liwiro's technical team, and operational playbooks for common scenarios.

Post-Implementation Support: Four months of dedicated technical support covering troubleshooting, optimization, monitoring assistance, and consultation for any infrastructure concerns.

Project Team Structure

Our implementation team comprises four specialized roles ensuring comprehensive expertise across all project phases:

IT Project Lead: Oversees entire project lifecycle, manages timelines and deliverables, coordinates with stakeholders, ensures quality standards, and serves as primary point of contact for Liwiro management.

Network Engineer: Designs and implements VPN infrastructure, configures routing and switching equipment, establishes connectivity with service providers, conducts network performance testing, and provides network security implementation.

Software Developer: Handles SMPP gateway installation and configuration, implements USSD TruRoute gateway, develops integration scripts and monitoring tools, conducts software testing, and provides application-level troubleshooting.

IT Support Specialist: Provides server preparation and configuration, assists with hardware and software installation, conducts connectivity testing, delivers end-user training, and manages the four-month support period.



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Implementation Timeline

Phase 1 - Planning & Coordination (Week 1): Initial stakeholder meetings, service provider engagement with TNM and Airtel, technical requirements documentation, and infrastructure assessment.

Phase 2 - Infrastructure Setup (Weeks 2-3): VPN configuration with service providers, LAN setup and router/switch configuration, server deployment and hardening, and preliminary connectivity testing.

Phase 3 - Gateway Implementation (Weeks 4-5): SMPP software installation and configuration, USSD TruRoute gateway deployment, integration with existing Liwiro systems, and comprehensive connectivity testing between all components.

Phase 4 - Testing & Validation (Week 6): End-to-end system testing, load and performance testing, failover scenario validation, and security assessment.

Phase 5 - Handover & Training (Weeks 7-8): Technical documentation delivery, staff training sessions, operational handover, and commencement of support period.

Phase 6 - Support Period (Months 3-6): Ongoing technical assistance, performance monitoring, optimization recommendations, and issue resolution.



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Financial Breakdown

Item	Description	Amount (MWK)
Professional Services		
IT Project Lead	Project management, stakeholder coordination, quality assurance (160 hours)	1,600,000
Network Engineer	VPN setup, network configuration, connectivity testing (200 hours)	2,000,000
Software Developer	Gateway installation, software configuration, integration (180 hours)	1,800,000
IT Support Specialist	Server setup, testing support, 4-month support period (160 hours + support)	1,200,000
Software & Licenses (Once-Off)		
SMPP Gateway Software	Enterprise license - one-time purchase, perpetual use	800,000
USSD TruRoute Gateway	Commercial license - one-time purchase, perpetual use	700,000
Service Provider Coordination		
TNM Integration Services	Liaison, documentation, testing coordination	150,000
Airtel Integration Services	Liaison, documentation, testing coordination	150,000



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Item	Description	Amount (MWK)
Project Management & Documentation		
Project Documentation	Technical documentation, user guides, operational manuals	100,000
Logistics & Contingency		
Logistics & Contingency	Travel, communications, unforeseen requirements	150,000
TOTAL		MWK 8,650,000

Note: All software licenses are one-time purchases with perpetual usage rights. No recurring annual license fees apply to these gateway solutions.



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Expected Benefits

Improved Performance: Direct connectivity eliminates third-party routing delays, reducing USSD session response times by up to 70% and significantly enhancing user experience.

Simplified Operations: Direct relationships with TNM and Airtel streamline troubleshooting, reduce mean time to resolution, and provide clearer visibility into system performance.

Enhanced Reliability: Dedicated infrastructure with redundant connectivity options ensures higher uptime and better service availability for Liwiro's customers.

Cost Efficiency: Elimination of third-party routing fees and reduced operational overhead from faster problem resolution creates long-term cost savings. With one-time software licenses, there are no recurring annual costs.

Scalability: The new architecture provides a solid foundation for future service expansion, additional channels, and increased transaction volumes without fundamental redesign.

Complete Control: Liwiro gains full visibility and control over their USSD and SMS infrastructure, enabling proactive monitoring and rapid response to issues.

Terms & Conditions

Payment Schedule: 40% upon contract signing and project commencement, 40% upon successful completion of gateway installation and connectivity testing, 20% upon final handover and acceptance.

Project Duration: Two months from commencement date to final handover and acceptance.

Support Period: Four months of dedicated technical support included, commencing from final handover date, covering remote assistance, troubleshooting, and consultation.



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Software Licenses: All software licenses are one-time purchases with perpetual usage rights. No recurring license fees or subscriptions required.

Documentation: All technical documentation, configuration guides, and operational procedures provided in electronic format with transfer of ownership to Liwiro Microcredit.

Training: Up to three full-day training sessions included for Liwiro technical staff covering system administration, troubleshooting procedures, and routine maintenance.

Warranty: 90-day warranty on all configuration and implementation work from final handover date, excluding issues arising from third-party service provider changes or unauthorized modifications.

Acceptance

We are confident this solution will transform Liwiro Microcredit's USSD and SMS infrastructure, delivering immediate performance improvements and establishing a robust platform for future growth. We look forward to partnering with you on this critical initiative.

Authorized Signature: _____

Name: Misheck Kamuloni

Position: CTO

Date: _____

Client Acceptance:

Authorized Signature: _____

Name: _____

Position: _____

Date: _____
